LINEAR MODELS APPLIED TO ECONOMETRICS

The influence of a master's degree on employability in France

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November 2024

SUMMARY

INTRODUCTION

«Graduates with higher education enjoy better job quality.»
(OECD, 2023)

In France, a higher education diploma is often seen as a stepping stone to specialized and high-value careers. In 2023, nearly 3 million students were enrolled in higher education, compared to 2.16 million in 2000, reflecting the growing importance of this qualification *(INSEE, 2024a)*. Higher education also provides protection against unemployment: the unemployment rate for graduates with a Bac+2 or higher was 5 %, compared to 13.3 % for those without a diploma *(INSEE, 2024b)*. However, this benefit varies by field. Some master's degrees have unemployment rates similar to those of vocational high school graduates *(INSEE, 2010)*. In terms of salary, graduates also enjoy a significant advantage: in 2016, the median net salary for a long-term graduate was €2,300, compared to €1,550 for a high school graduate *(INSEE, 2018)*. However, disparities persist based on the school attended and the chosen specialization. A study by *KEDGE (2023)* highlights significant salary gaps depending on the industry and the institution.

Certain fields, such as artificial intelligence, information technology, and environmental project management, see high demand for advanced technical skills. Internationally, more than 60% of the workforce specialized in AI holds a higher education diploma *(OECD, 2023)*. Furthermore, the type of program pursued – whether a research master's, professional master's, or MBA – shapes employer perceptions, particularly regarding practical application and innovation. apprenticeships programs, in particular, are highly valued by recruiters, as they combine theory with practical experience *(APEC, 2018)*. Given the increased competition among graduates and the rapid evolution of required skills, it becomes crucial for students to understand the impact of their academic choices on their employment prospects and career paths.

What impact do the characteristics of disciplines and diplomas have on the employment, job stability, and working conditions of young graduates? In the first section, we will examine the general context of employability for young graduates in France, exploring indicators, the impact of master's degrees, and public and private levers. In the second section, we will focus on the factors influencing their professional integration and job quality while identifying current challenges and trends.

I) The state of the labor market in France for young graduates : how growing insertion rates highlight the effect public policies and private initiatives

"The diploma is a major asset in the labor market" (Nauze-Fichet, 2002). However, this asset is not uniform for all young graduates. Understanding the challenges related to their employability requires examining current indicators and evaluating the impact of degrees and fields of study on career trajectories.

A) Employability of Young Graduates in France: Indicators, Impact of a Master's Degree, and Perspectives

"The term 'employability' refers to transferable skills and qualifications that enhance an individual's ability to take advantage of available education and training opportunities, secure decent work, retain it, progress in the company or by changing jobs, and adapt to evolving technologies and labor market conditions" (International Labour Organization, *Recommendation No. 195 on Human Resources Development*, 2004).

1. Employability: A Broad Term Defining an Individual's Capability to Be Employed

The concept of employability, which originated in Great Britain during the Industrial Revolution, initially differentiated individuals capable of integrating into society from the poorest relying on public assistance. In the 1930s, it was adapted in the United States to guide recovery plans following the 1929 economic crisis, identifying profiles deemed fit for work. After World War II, in a favorable economic climate, the term evolved to measure the employment gap for disadvantaged groups. It transitioned from a medical approach to individual abilities to a focus on overall attractiveness to employers, eventually becoming the current tripartite interaction between individuals, the state, and the labor market (Ben Hassen & Hofaidhllaoui, 2012). Employability is based on four main dimensions: finding a job, retaining it, progressing within it, and securing another one. Its development is influenced by several types of variables. Individual variables include a broad range of factors, such as skills, professional qualifications, experiences, and personal traits like motivation, resilience, or self-esteem. Other factors, such as age, gender, and health status, also play a role, as does the ability to position oneself and adapt to the labor market. Managerial and structural variables pertain to skills management within organizations. This includes understanding the current and future skills of employees, organizing work to foster skill development, empowering employees in their career advancement, and assessing skills acquired through certification systems. Finally, Environmental variables relate to the external context, such as legislation, social policies, union agreements, economic conditions, and educational and institutional infrastructures *(Ben Hassen & Hofaidhllaoui, 2012)*.

The term "employability" must be used cautiously, as it initially seems to focus on the supply of workers, whereas the demand side, influenced by businesses and public policies, may be just as significant in determining its overall level (Gazier, 2017). This distinction highlights two types of employability: relative employability, which evaluates an individual's productivity on a personal level, and absolute employability, which takes into account labor market conditions and the broader economic context (Ben Hassen & Hofaidhllaoui, 2012). The challenge of the concept of employability lies in understanding the insertion and employment rates of a given population. Examining its components, such as the individual variables mentioned above, helps adapt public policies and adjust strategies by acknowledging the vast scope of employability levers: "the knowledge or expertise provided through a learning process, often long and validated by a degree, is broadened to encompass everything that attests to the student's ability to adapt, evolve, and effectively mobilize the appropriate resources to address the challenges posed by the rapid evolution of work and jobs." (Béduwé & Mora, 2017).

Professional integration refers to the process by which young graduates secure stable employment, marking their transition from student to active worker. Vernières (1997) defines this stage as young entrants achieving a stable position in the labor market (INSEE, 2004). The insertion rate measures the proportion of graduates who are employed out of all graduates in the labor market, whether employed or unemployed (Ministry of Higher Education, 2018). This rate is a key metric for assessing the effectiveness of the education system in preparing individuals for employment. Stable employment is another key indicator, reflecting the proportion of graduates with permanent contracts ("CDI" in France), positions in the public sector, or as self-employed individuals, highlighting the sustainability of professional integration (Ministry of Higher Education, 2018). Entry-level salary, an indicator of a degree's market attractiveness, varies based on the field of study, region, and company. According to recent studies, such as one conducted by WTW, salaries for young graduates increased in 2023, particularly for those with a Bac+5 qualification (business schools, engineering schools, and prestigious institutions). These profiles, now more sought-after and demanding, saw their median salary rise significantly (+7 %), reaching €43,000 last year (ISCOD, 2024). The graduate unemployment rate measures the market's ability to absorb new entrants. In 2021, 9.8 % of young graduates who had completed their studies within the past 1–4 years were unemployed (INJEP, 2023). Finally, the time required to secure the first job is an indicator of how long it takes to find employment. About 40% of young people take more than a year to secure their first stable job (INSEE, 2009).

2. Masters in France: levers for successful professional integration?

Historically, diplomas have always been a social marker and a selection criterion for employers. However, with the intensification of competition on the job market since the 1990s, due in particular to the massification of higher education between 1960 and 1990, their importance has continued to grow (Benhenda & Dufour, 2015). In France, the proportion of young working people with a bachelor's degree or higher rose from 44 % in 1990 to 67 % in 2001. This increase was also observed for higher qualifications: from 28 % to 45 % for post-bac studies over the same period (INSEE, 2022). According to a study published by Céreq, 85 % of Master's graduates are in employment three years after graduation. Of these, 75 % have a stable contract, and their median net salary is €2,170 per month. The specialization of the degree has a direct influence on these results: science, engineering and management courses show high insertion rates, while graduates in the humanities or arts encounter more difficulties (Céreq, 2024). However, the French labor market has also seen an increase in cases of "downgrading", where graduates take jobs for which they are overqualified. This reflects an imbalance between the growing supply of skilled labor and the demand for qualified jobs. This situation is exacerbated by economic conditions and structural changes in the labor market. Downgrading can affect employee motivation and productivity, while posing challenges for optimizing the skills available on the labor market (Céreq, 2014). The Céreq study identifies seven insertion trajectories, ranging from rapid, stable paths to precarious situations. Graduates in engineering, IT and healthcare enjoy the best prospects, with employment rates close to 95 %. Conversely, literary or artistic paths often lead to precarious employment or jobs outside their field of specialization, with lower salaries (Céreq, 2024).

The report from the French Ministry of Higher Education (MESR, 2023) points out that 93 % of Master's graduates are in employment 30 months after leaving university. This high score can be explained by the widespread use of internships, the growing popularity of apprenticeships, Masters programs, and the strengthening of links with the professional world. Professionalizing Masters programs, including internships and apprenticeships programs, play a key role in the professional integration of graduates. According to (MESR, 2023), 93 % of Master's graduates are in employment thirty months after graduation, thanks to schemes such as sandwich courses and internships, which reinforce practical skills and facilitate integration into the professional world. Céreq (2024) shows that these courses offer better integration than research Masters, except for those pursuing a doctorate. The Céreq Échanges study (2024) confirms that apprenticeships students benefit from a higher hiring rate, as companies value their academic and practical experience. The French Ministry of Higher Education continues to encourage these schemes to align university training with market needs, thereby enhancing the employability of graduates.

University Masters programs in France are powerful levers for professional integration, but their impact varies according to the discipline, educational background and socio-economic circumstances of graduates. Current initiatives, notably sandwich courses and internships, are helping to boost graduate employability, but particular attention needs to be paid to social and disciplinary disparities.

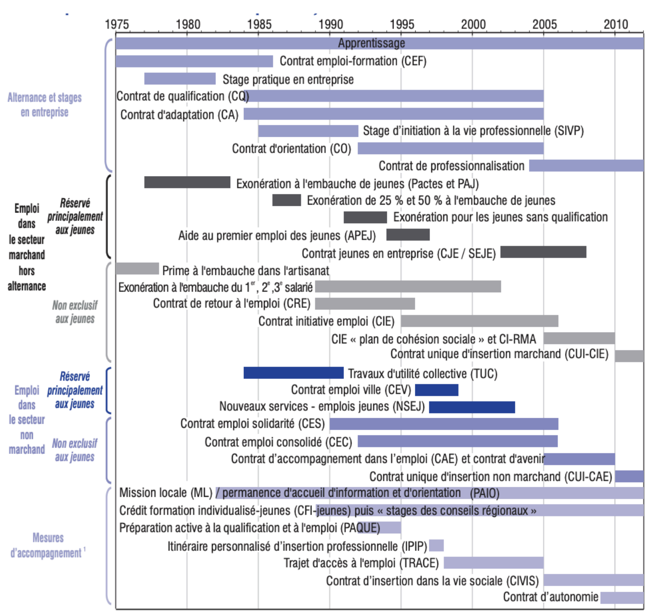
B) Levers for Professional Integration and the Impact of Higher Education Privatization on

After exploring the fundamental concepts and key indicators related to employability, we will now examine the various levers that can optimize the integration of graduates into the labor market. These levers, whether institutional, educational, or linked to corporate engagement, play a central role in reducing inequalities in access to employment and aligning skills with economic needs.

1. **The Impact of Public Policies on Graduate Employability: An Increase in Programs to Facilitate Young People's Access to the Job Market.**

Several initiatives aim to promote youth employment, focusing on two main objectives: reducing barriers to entry and enhancing skills. The ARPE, designed for graduates under 28 who received a scholarship during their final year of study, provides financial support to ease the transition between education and the first job (Ministry of Higher Education, 2017). Other initiatives include the Youth Engagement Contract (CEJ), "second chance" programs, as well as apprenticeships and professional training contracts. Since 2020, the *1 young person, 1 solution* plan, backed by over €9 billion, combines hiring incentives, training, support, and financial assistance to address the specific needs of every young individual (Ministry of Labor, 2023). Programs such as subsidized employment contracts (CAE and CIE) also encourage the hiring of inexperienced young people by offering wage subsidies to employers. These measures provide beneficiaries with valuable initial work experience, helping them integrate into the workforce (INSEE, 2011). Finally, programs like the Personal Training Account (CPF) and the Investment Plan for Skills (PIC) boost youth employability by offering continuous training and development opportunities, enabling them to meet the demands of a constantly evolving job market.

The graph below illustrates the evolution of public policies implemented to support employability in France :



Les politiques d'accès à l'emploi en faveur des jeunes - INSEE, 2011

Numerous initiatives have been deployed to facilitate youth access to the job market, but it is crucial to evaluate their outcomes. According to INSEE, in 2010, nearly 24% of jobs held by young people under 26 were supported by public assistance, representing about 665,000 positions (INSEE, 2011). An analysis of beneficiaries of subsidized contracts, based on data from the 2008 Dares Panel, reveals a complex relationship between these programs and their impact on the labor market. The profiles of beneficiaries vary: some, far from employment, require intensive support, while others, closer to the job market, access more integrated contracts. Employment outcomes six months after the contract ends depend on factors such as the duration of the program, the support received, and the characteristics of the employer. Overall, beneficiaries have a positive perception of these contracts, highlighting their role in strengthening their employability (INSEE, 2015).

Since the 1990s, public policies have intensified individualized support programs, combining advice, tailored training, and job search assistance. These initiatives aim to overcome the specific barriers faced by young people. Evaluations show that, in the private sector, subsidized contracts significantly increase the rate of non-subsidized employment two years after their implementation. However, their effectiveness remains limited, or even negative, in the non-profit sector (Card, Kluve, and Weber, 2010; Kluve, 2010). These results underscore the importance of targeted programs that combine financial support, professional qualification, and career-building assistance.

2. How the private sector is taking over education

We previously discussed the importance of individual qualifications in enhancing a person’s value in the labor market. However, this significance should not overshadow the role of the market itself, as it is responsible for structuring demand, either favoring or sidelining certain profiles. A striking example: “Since 2008, the employment rate of individuals with a bachelor's degree or higher has remained remarkably stable, while it has declined for lower education levels, with sharper drops for those with fewer qualifications. This suggests more of an upgrading (an increase in qualification levels) than a polarization” (Askenazy, Behaghel, Laouenan & Meurs, 2019). Since the role of employers is crucial, we will now focus on companies’ recruitment policies, particularly their relationship with training. While continuing education emerged in the 1950s, it wasn’t until the 1970s that it was addressed in specific legal frameworks, and in the 2000s that it underwent a transformation to better align with the needs of individual companies (Benadbeljalil, 2023). In 2020, the OECD defined it as “training undertaken to acquire the knowledge or skills required in current or targeted roles, increase earnings, improve career prospects within the sector one works in or another sector, or more generally, expand promotion opportunities” (OECD, 2020). Thus, continuing education is not only a means for employees to enhance their ability to retain a job but also to find new opportunities, significantly impacting their employability.

Over the past three decades, professional training has given rise to a new concept: corporate universities. France now has nearly a hundred of these, primarily within multinational groups. These institutions can train diverse populations (technical staff, high-potential employees, new hires) and may be linked to various departments (general management, sales, HR). They adapt to the company's needs, acting as showcases for expertise, tools for career advancement, and mechanisms to align staff with the organization's culture, values, or operational requirements (Salmon, 2014).

Beyond corporate universities, there is now a clear trend of private companies integrating themselves into the education sector. For instance, the recent acquisition of the École Supérieure de Journalisme (ESJ) by Vincent Bolloré, Rodolphe Saadé, Bernard Arnault, and the Dassault family highlights this shift. Additionally, there are close partnerships between schools and private companies, such as the one between Albert School Marseille and CMA CGM. Parallelly, companies are increasingly developing their own training centers, such as the CMA CGM Academy or Louis Vuitton's Institut des Métiers d'Excellence. The latter offers apprenticeship programs focused on creative and artisanal professions to preserve and pass on unique expertise. Through these initiatives, major corporations are no longer merely waiting for suitable candidates or training their employees; they are actively shaping the educational landscape.

Private institutions, particularly business and engineering schools, have become increasingly attractive by promising strong professionalization and quick access to employment. The programs offered by these schools are often better aligned with labor market needs thanks to partnerships with companies, apprenticeships programs, and specialized training. Graduates from these institutions typically enjoy better job placement, with higher employment rates and entry-level salaries compared to their peers from public institutions. However, access to these programs remains expensive, with tuition fees often exceeding €10,000 per year. This financial barrier fosters elitism, as only students from affluent backgrounds can afford these programs. The impact on employability is therefore a double-edged sword: while a degree from a private school enhances the likelihood of securing quality employment, it remains accessible to only a minority of students who can bear the cost (Lauwerier, 2022).

II) Analysis of the determinants of the professional integration and job quality of young graduates in France: Issues and policy recommendations

"Young graduates do not all have equal access to stable employment. Public policies must take this diversity into account to ensure fair access to job security." (Ministry of Labour, Report on the professional integration of young people, 2023).

A) Database: descriptive statistics and illustrations

Now that we have identified the various levers of public policies and professional institutions, it is crucial to examine concrete data to better understand the reality of young graduates' *professional integration. To this end, we will analyze a representative sample of Master's degree holders from diverse schools and disciplines.*

1. Description and Presentation of the Dataset

The dataset used for this analysis provides comprehensive insights into the professional integration of Master's graduates in France, with a particular focus on employment stability, job quality, and the socio-economic factors influencing early career trajectories. Comprising 7,346 observations across 19 variables, it serves as a robust foundation for understanding the determinants of stable employment among young graduates. The data originates from a national survey conducted by the Ministry of Higher Education and Research, targeting graduates from universities and equivalent institutions. This survey, conducted 30 months post-graduation, evaluates various dimensions of professional integration, including insertion rates, job stability, and salary levels. To ensure reliability, the dataset includes only institutions with a response rate above 30% and a minimum number of respondents.

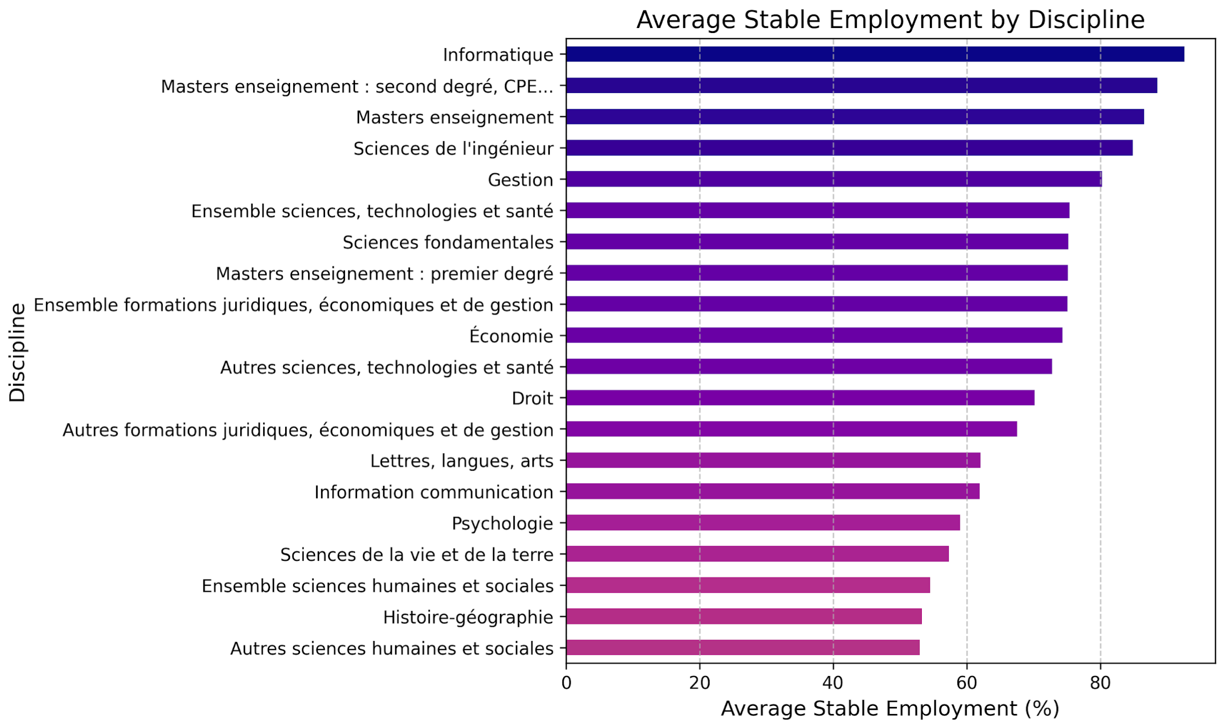
The dataset captures key demographic and institutional variables, such as the year of graduation, type of degree, academic field, and institutional affiliation. These variables enable comparisons across institutions and disciplines, as well as temporal analysis of labor market outcomes. Indicators of professional integration, such as the proportion of graduates employed, the shareholding stable positions (e.g., permanent contracts), and the percentage working in managerial or intermediate professional roles, provide critical measures of job quality. Additionally, the dataset details the prevalence of full-time employment, offering insights into the intensity of labor market participation.

Income measures are also well-documented, including the median net salary for full-time positions and the estimated annual gross salary. These variables are contextualized with regional benchmarks, such as the median monthly salary for young managers, enabling an analysis of local labor market disparities. The socio-economic context is further enriched with variables like the proportion of graduates who received scholarships during their studies and the share of graduates employed outside their home region, reflecting geographic mobility and economic background. Gender dynamics are also captured, allowing for gender-specific analyses of employment outcomes.

Finally, discipline-specific variables, such as the relative weight of each field within the graduate population, provide a nuanced understanding of labor market dynamics across academic disciplines. Supplementary data from INSEE, including regional unemployment rates and national discipline-level trends, enhances the richness of this dataset. Overall, this data allows for a detailed analysis of the factors influencing job stability and quality, setting the stage for the econometric modeling and empirical exploration presented in the following sections.

2. Exploring Disparities in Job Stability Across Academic Fields

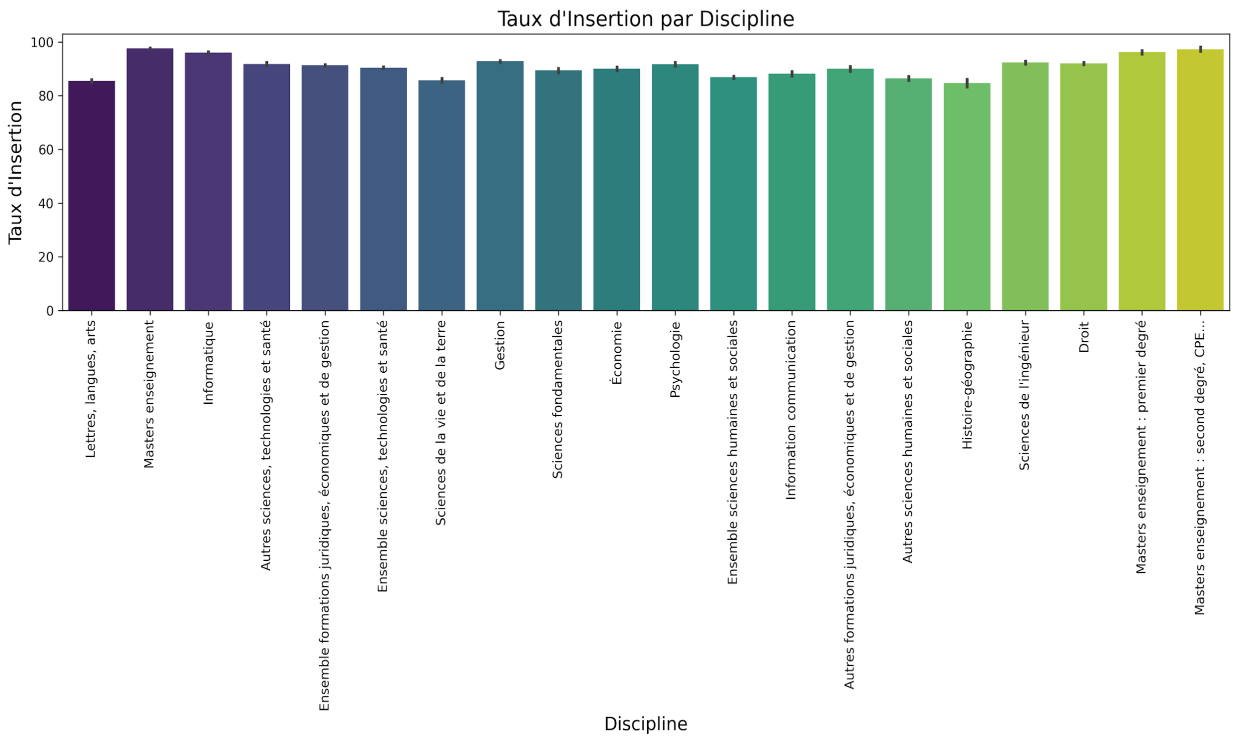
Throughout this study, certain statistics particularly caught our attention. On average, 47.01% of graduates in the humanities and social sciences occupy precarious jobs, while 92.59% of computer science graduates have stable employment. These figures highlight significant disparities between different academic fields. Disciplines like computer science seem to offer much higher rates of stable employment compared to fields like the humanities and social sciences. This difference raises important questions about the factors underlying these gaps.



To better understand these disparities and identify the key elements that determine access to stable employment, it is relevant to undertake a more in-depth analysis using regression. This method will allow us to statistically examine the relationships between variables such as professional integration rates, degree level, field of study, and job stability. Such an approach will provide a clearer understanding of the underlying determinants and can serve as a basis for policy recommendations aimed at reducing these inequalities.

B) Using econometrics to inform policy decisions on education and employment

1.The impact of discipline, institution and other factors on the entry rate and job stability of young graduates



Initially, we developed a simple regression model:

$Y = \beta 0 + \beta 1X + \epsilon$

Where Y: Professional integration rate of young graduates.
And X: Discipline of the master's degree pursued.

We chose to set up a model that explains the professional integration rate of young graduates based on the discipline pursued, as it plays a decisive role in accessing the job market and the quality of employment. Indeed, the field of study is a relevant explanatory variable for understanding differences in professional integration and formulating recommendations aimed at improving the transition from education to employment.

This model allowed us to obtain the following results:

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                            OLS Regression Results
==============================================================================
Dep. Variable:        taux_dinsertion   R-squared:                       0.336
Model:                            OLS   Adj. R-squared:                  0.335
Method:                 Least Squares   F-statistic:                     195.3
Date:                Sun, 17 Nov 2024   Prob (F-statistic):               0.00
Time:                        21:42:03   Log-Likelihood:                -21839.
No. Observations:                7346   AIC:                         4.372e+04
Df Residuals:                    7326   BIC:                         4.386e+04
Df Model:                          19
Covariance Type:            nonrobust
==============================================================================================================================================
                                                                                 coef    std err          t      P>|t|      [0.025      0.975]
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Intercept                                                                     90.0061      0.371    242.615      0.000      89.279      90.733
discipline[T.Autres sciences humaines et sociales]                            -3.5717      0.464     -7.703      0.000      -4.481      -2.663
discipline[T.Autres sciences, technologies et santé]                           1.8023      0.492      3.660      0.000       0.837       2.768
discipline[T.Droit]                                                            1.9653      0.432      4.546      0.000       1.118       2.813
discipline[T.Ensemble formations juridiques, économiques et de gestion]        1.3289      0.401      3.317      0.001       0.544       2.114
discipline[T.Ensemble sciences humaines et sociales]                          -3.1474      0.410     -7.670      0.000      -3.952      -2.343
discipline[T.Ensemble sciences, technologies et santé]                         0.4500      0.405      1.112      0.266      -0.344       1.244
discipline[T.Gestion]                                                          2.7966      0.407      6.864      0.000       1.998       3.595
discipline[T.Histoire-géographie]                                             -5.3506      0.622     -8.602      0.000      -6.570      -4.131
discipline[T.Information communication]                                       -1.8508      0.497     -3.728      0.000      -2.824      -0.877
discipline[T.Informatique]                                                     5.9681      0.469     12.721      0.000       5.048       6.888
discipline[T.Lettres, langues, arts]                                          -4.5183      0.445    -10.152      0.000      -5.391      -3.646
discipline[T.Masters enseignement]                                             7.6375      0.421     18.133      0.000       6.812       8.463
discipline[T.Masters enseignement : premier degré]                             6.2082      0.969      6.407      0.000       4.309       8.108
discipline[T.Masters enseignement : second degré, CPE...]                      7.3016      1.000      7.300      0.000       5.341       9.262
discipline[T.Psychologie]                                                      1.6673      0.502      3.321      0.001       0.683       2.652
discipline[T.Sciences de l'ingénieur]                                          2.3546      0.467      5.046      0.000       1.440       3.269
discipline[T.Sciences de la vie et de la terre]                               -4.2667      0.443     -9.621      0.000      -5.136      -3.397
discipline[T.Sciences fondamentales]                                          -0.5879      0.495     -1.187      0.235      -1.559       0.383
discipline[T.Économie]                                                         0.0223      0.478      0.047      0.963      -0.915       0.960
==============================================================================
Omnibus:                      649.342   Durbin-Watson:                   1.684
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              991.576
Skew:                          -0.679   Prob(JB):                    4.81e-216
Kurtosis:                       4.180   Cond. No.                         32.1
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The results of the analysis confirm that the field of study is a determining factor in the professional integration of young graduates. Certain fields, such as computer science or teaching professions, positively influence the integration rate, with a significant variation of 5.9681%, indicating a favorable integration into the labor market. On the other hand, disciplines such as literature or the humanities have a negative impact on the integration rate, suggesting that the professional integration of graduates from these fields is more complex. These observations highlight the disparity in employment prospects depending on the field of study.

Although the field of study plays a significant role in the professional integration of young graduates, it does not fully explain the variations observed in the integration rate, as evidenced by the R^2 coefficient of 0.336. Indeed, nearly 66.4% of the variability in the integration rate remains unexplained by this model. This suggests that other factors, not considered in this simplified approach, could also influence professional integration, such as the economic context, the specific skills of individuals, as well as social and geographical variables. It would therefore be advisable to develop more complex econometric models that incorporate additional explanatory variables in order to better capture the underlying dynamics and provide a more accurate analysis of the phenomenon.

The development of a more complex model appears necessary to refine the estimation of the professional integration rate of young graduates. Thus, we enhanced our simple regression model by adding an additional explanatory variable: the graduates' institution of origin. This multivariate regression model allowed us to obtain the following results:

OLS Regression Results

Dep. Variable:	y	R-squared:	0.582
Model:	OLS	Adj. R-squared:	0.577
Method:	Least Squares	F-statistic:	100.0
Date:	Sun, 17 Nov 2024	Prob (F-statistic):	0.00
Time:	23:14:24	Log-Likelihood:	-27334.
No. Observations:	7346	AIC:	5.487e+04
Df Residuals:	7244	BIC:	5.558e+04
Df Model:	101		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	64.8479	1.017	63.736	0.000	62.853	66.842
x1[T.Autres sciences humaines et sociales]	-13.2549	1.016	-13.051	0.000	-15.246	-11.264
x1[T.Autres sciences, technologies et santé]	5.2844	1.084	4.873	0.000	3.158	7.410
x1[T.Droit]	2.2770	0.945	2.410	0.016	0.425	4.129
x1[T.Ensemble formations juridiques, économiques et de gestion]	7.2689	0.878	8.277	0.000	5.547	8.991
x1[T.Ensemble sciences humaines et sociales]	-12.7852	0.900	-14.200	0.000	-14.550	-11.020
x1[T.Ensemble sciences, technologies et santé]	7.6858	0.893	8.606	0.000	5.935	9.436
x1[T.Gestion]	12.7315	0.891	14.293	0.000	10.985	14.478
x1[T.Histoire-géographie]	-13.7681	1.364	-10.097	0.000	-16.441	-11.095
x1[T.Information communication]	-6.0263	1.113	-5.415	0.000	-8.208	-3.845
x1[T.Informatique]	24.1200	1.036	23.285	0.000	22.089	26.151
x1[T.Lettres, langues, arts]	-4.1986	0.978	-4.292	0.000	-6.116	-2.281
x1[T.Masters enseignement]	19.4794	0.922	21.121	0.000	17.671	21.287
x1[T.Masters enseignement : premier degré]	8.1860	2.080	3.935	0.000	4.108	12.264
x1[T.Masters enseignement : second degré, CPE...]	22.4229	2.143	10.465	0.000	18.223	26.623
x1[T.Psychologie]	-7.5195	1.107	-6.794	0.000	-9.689	-5.350
x1[T.Sciences de l'ingénieur]	16.1937	1.029	15.734	0.000	14.176	18.211
x2[T.Corse Pasquale Paoli]	-15.2524	3.618	-4.216	0.000	-22.345	-8.160
x2[T.Côte d'Azur]	10.9057	1.966	5.548	0.000	7.053	14.759
x2[T.Paris 2 - Panthéon Assas]	4.8812	1.464	3.334	0.001	2.012	7.751
x2[T.Paris 3 - Sorbonne Nouvelle]	-1.6060	1.497	-1.073	0.284	-4.541	1.329
x2[T.Paris 4 - Sorbonne]	11.9039	1.950	6.104	0.000	8.081	15.727

The inclusion of the "institution" variable in our regression model had a notable impact on the coefficients of the disciplines. Specifically, the coefficient associated with the computer science discipline increased significantly, reaching 24.12, suggesting that belonging to this field, when accounting for the institution, is strongly linked to a higher integration rate. On the other hand, the introduction of this variable amplified the negative effect of humanities disciplines, with the coefficient now even lower, indicating that graduates from these fields face more pronounced integration difficulties, even when the institution is taken into account. It also emerges that the university of origin has a significant impact on the integration rate. Graduates from prestigious universities, such as the Sorbonne, in renowned fields like engineering and medicine, benefit from a much more favorable integration rate. In contrast, those from less recognized universities, such as Sorbonne Nouvelle, show a negative association with the integration rate, as the coefficient is negative. Furthermore, the significance test indicates that the P-value is less than 0.05 ($P > |t| = 0$), confirming that the results obtained are statistically significant and that the variables "discipline" and "institution" have a real impact on the professional integration rate of young graduates.

After analyzing the impact of disciplines and institutions on the integration rate of young graduates, it appears that these factors play a decisive role in the initial success of their entry into the job market. However, professional integration is not limited to this first phase. Job stability is also a key indicator of the professional success of young graduates. In this perspective, we expand our analysis by focusing on the determinants of job stability. This approach aims to identify the factors, beyond initial integration, that influence job sustainability within this population. We will integrate relevant explanatory variables into our model, such as estimated gross annual salary, the proportion of graduates with full-time jobs, the share of graduates holding managerial positions, whether the institution belongs to the Paris academic district, as well as whether the discipline has a high integration rate. The goal is to provide an in-depth analysis of the conditions that promote long-term job stability for young graduates, while taking into account academic, institutional, and professional specifics that may influence this dynamic.

Our multiple regression model is represented by the following linear equation:

$Y = \beta0 + \beta1 \cdot X1 + \beta2 \cdot X2 + \beta3 \cdot X3 + \beta4 \cdot X4 + \beta5 \cdot X5 + \epsilon$

β0: Intercept (or constant) of the regression.
β1: Coefficient associated with 'salaire_brut_annuel_estime' (the estimated gross annual salary).
β2: Coefficient associated with 'emplois_a_temps_plein' (the percentage of graduates working full-time).
β3: Coefficient associated with 'emplois_cadre' (the percentage of graduates holding managerial positions).
β4: Coefficient associated with 'Etablissement_parisien' (binary indicator for Paris-based institutions: 1 if the institution is in Paris, 0 otherwise).
β5: Binary indicator for top 5 disciplines in terms of integration rate by field: 1 if the discipline is part of the top 5, 0 otherwise).
Top 5 disciplines: Teaching Masters: Primary level, Teaching Masters, Computer Science, Management.
ϵ: Random error (error term in the model).

The results obtained are as follows :

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.592
Model:                            OLS   Adj. R-squared:                  0.592
Method:                 Least Squares   F-statistic:                     2132.
Date:                Mon, 18 Nov 2024   Prob (F-statistic):               0.00
Time:                        00:32:38   Log-Likelihood:                -27246.
No. Observations:                7346   AIC:                         5.450e+04
Df Residuals:                    7340   BIC:                         5.454e+04
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    -39.8129      1.418    -28.075      0.000     -42.593     -37.033
x1             0.0020   4.66e-05     43.825      0.000       0.002       0.002
x2             0.4814      0.019     25.479      0.000       0.444       0.518
x3             0.0661      0.009      7.658      0.000       0.049       0.083
x4            -8.0533      0.424    -18.999      0.000      -8.884      -7.222
x5            20.2911      0.490     41.410      0.000      19.331      21.252
==============================================================================
Omnibus:                      186.177   Durbin-Watson:                   1.725
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              212.817
Skew:                          -0.354   Prob(JB):                     6.13e-47
Kurtosis:                       3.439   Cond. No.                     3.71e+05
==============================================================================
```

The integration of several explanatory variables into our model resulted in a coefficient of determination R^2 of 0.592, indicating that 59.2% of the variability in job stability is explained by the included variables. This shows that the model offers a satisfactory explanatory power, although a significant portion of the variance (40.8%) remains unexplained. The intercept is estimated at -39.8129, representing the theoretical value of job stability when all explanatory variables are zero. Although this term does not have an interpretable meaning in this context, it serves as a baseline to evaluate the effects of the other coefficients.

The coefficient β3 = 0.0661 reveals that a 1% increase in the proportion of graduates holding managerial positions is associated with an increase of 0.0661 points in job stability, all else being equal. This positive relationship suggests that managerial positions, often accompanied by more stable contracts and better conditions, contribute to greater job sustainability. In contrast, the coefficient β4 = -8.0533 shows that, all else being equal, belonging to a Paris-based institution is associated with a decrease of 8.05 points in job stability. This observation, although counterintuitive, could be explained by intense competition in the job market within the highly sought-after sectors of the Paris region.

Finally, the coefficient β5 = 20.2911 highlights the positive effect of belonging to a discipline with a high integration rate: graduates from these disciplines experience an increase of 20.29 points in job stability. This result underscores the importance of disciplinary choices, suggesting that high-demand sectors such as computer science or engineering not only promote better professional integration but also greater job sustainability.

2. The *Modus Operandi* of Public Policies to Improve the Professional Integration of Young Graduates and Reduce Sectoral Disparities

Strengthening Professionalization and Apprenticeship Programs

To sustainably support high employment rates, it is essential to promote policies encouraging apprenticeships and internships programs while enhancing collaboration between private companies and the education system. Professional master's degrees that include mandatory internships or apprenticeships options play a crucial role in integrating young graduates into the workforce. In 2023, the number of apprentices reached 1,021,500, marking a 7.1% increase compared to 2022 (*Alternance professionnelle*, 2024). To build on this progress, higher education institutions should establish minimum quotas for internships or apprenticeships placements within their programs. Additionally, increasing corporate involvement in education, as exemplified by initiatives such as the CMA CGM Academy or Louis Vuitton's Institut des Métiers d'Excellence, can help align academic training more closely with labor market needs. Such measures would bridge the gap between acquired skills and employer expectations, enhancing the employability and attractiveness of graduates.

Addressing Supply-Demand Mismatches: Orientation Campaigns and Targeted Partnerships

Improving employment rates in struggling sectors requires a dual approach. On the supply side, the key challenge is the mismatch between the qualifications of young graduates and labor market demands. Fields like the humanities often face precarious employment prospects, with stable employment rates lower in disciplines such as history and geography (53.29%), psychology (59.04%), and communication (61.98%) compared to more technical fields like engineering (84.86%) and computer science (92.59%). Early intervention through orientation campaigns, discovery internships, and targeted guidance towards high-demand careers in areas such as engineering, IT, and management could help redirect students from oversaturated fields toward those offering more opportunities. Furthermore, closer collaboration between academic institutions and employers should ensure that students acquire skills directly applicable in professional contexts.

On the demand side, aligning investment and partnership strategies with France's academic and professional strengths is essential. Renowned for its excellence in engineering, research, and advanced industries, France should prioritize creating opportunities that match its talent pool. Public policies can direct investments toward growing sectors such as green technologies and artificial intelligence, while fostering public-private partnerships to generate jobs aligned with national expertise. In oversaturated fields like history, literature, and psychology, innovative public policies can create new opportunities. For instance, history and literature graduates could contribute to cultural heritage digitization projects or educational tourism initiatives, while psychology graduates could take on expanded roles in community mental health programs or as part of school systems. These measures would not only reduce saturation in these fields but also amplify their societal contributions.

Facilitating International Mobility for Graduates in Oversaturated Fields

When domestic strategies fall short, particularly in saturated academic fields, exporting surplus talent emerges as a viable solution. Limited domestic demand in disciplines like history or literature often reduces career opportunities for graduates. Encouraging international mobility could effectively address this challenge. Drawing inspiration from India's IT sector, which has positioned itself as a global "hub of digital expertise," France could adopt a similar model for exporting intellectual talent. India's IT industry exemplifies the success of such an approach. With 4 million IT professionals, the sector generated $147 billion in export revenue in 2020, accounting for 45% of India's service exports and contributing 8% to its GDP. Moreover, it dominates 55% of the global IT outsourcing market (*CCIFI, 2022)*. This strategic export of digital skills has solidified India's position as a global leader in IT services.France could leverage this model by fostering bilateral agreements with countries that value expertise in culture, education, or research. Graduates in history, humanities, or other specialized fields possess strong analytical and communication skills, making them ideal candidates for roles in international organizations, cultural institutions, or academic settings. Collaborations with EU member states or regions seeking expertise in education and culture could further expand opportunities for young French professionals.This approach not only alleviates domestic labor market pressures but also enhances France's intellectual and cultural influence on the global stage, mirroring India's successful integration into the global economy through its IT sector.

By combining a professionalized education system tailored to employer needs, targeted orientation policies to guide students toward growth sectors, strategic investments accompanied by robust public-private partnerships, and a proactive approach to international mobility for surplus talent, France can establish a comprehensive strategy for optimizing employment. This integrated framework would strengthen the alignment between graduates' skills and market expectations while creating opportunities in emerging sectors. Such a strategy would not only sustain high employment rates but also reinforce France’s global competitiveness and influence.

CONCLUSION

This study underlines the complex interaction of academic disciplines, institutional prestige, public policies, and market demand regarding the employability, job stability, and working conditions of young graduates in France. The findings bring out a nuanced reality: on one hand, higher education remains a critical lever for social mobility and economic stability; but on the other hand, its benefits are distributed in a very uneven way, with marked disparities according to fields of study, institutions, and regions. This calls for a holistic approach to reforming education-to-employment pathways. Democratization through mass higher education has increased the absolute numbers this group enters the labor market from their studies, which grew very fast. For those areas marked by saturation-humanities and social sciences-the competition has become fierce, and precarious employment is always a risk. By contrast, STEM and management fields secure the graduates in a leading position, considering their higher job placement rates, job security, and rational salaries. This situation simply reflects a labor market that is increasingly in need of technical specialized competencies at a much faster rate than general knowledge is supplied.

Such elements as institutional prestige and integration of professionalized education models, such as apprenticeships programs, internships, and corporate partnerships, emerge as key in enhancing graduate employability. Graduates coming from elite institutions with professionalized programs would present larger networks, practical experience, and better alignment with market needs and therefore enjoy improved job integration and job quality. On the other hand, high private education costs are accentuating socio-economic inequalities and rendering such opportunities inaccessible for students coming from less privileged backgrounds. Public policy plays an indispensable role in tackling disparities and providing opportunities for all to access quality employment. Interventions Such as providing subsidies for financially disadvantaged students, investing in emerging industries such as AI and green technologies, promoting vocational training, among others-should be pursued. On the other hand, early professional orientation and counseling toward highly demanded disciplines could help bridge the gap between the aspirations of further education and real demand on the labor market.

The exportation of talent by means of international mobility serves as a way out for fields where domestic saturation is often envisioned. Building on France's tradition of greatness in the cultural and intellectual domains, partnerships established with global institutions and international organizations may provide outlets for graduates in the humanities and arts. Such a measure would relieve the job market at home while reinforcing France's influence abroad. Such challenges of graduate employability require an integrated multi-stakeholder approach. Indeed, it is the policymakers, education providers, and employers who must collaborate in developing a sustainable ecosystem that aligns academic output with economic needs, yet maintains the wider value of studying diverse fields to society. This will involve balancing the short-term requirements of the labor market against long-term development of adaptable, critical thinkers competent to negotiate an increasingly changing world.

Conclusively, there exist genuine disciplinary and diploma influences on young graduates' professional routes; however, these interact with greater economic, institutional, and policy contexts. Embracing globalization, promoting innovation, and addressing the disparities should facilitate France in building a system from education to employment that would serve individual aspirations and reinforce the socio-economic fabric in an increasingly interdependent world. Achieving this vision requires continuous adaptation, foresight, and a commitment to equity-principles that ensure that higher education continues to create a catalyst for opportunity and progress.

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